

CrabTransformer Training Report

Scenario: REAL | Loss: Tweedie | Lag: 0 Years | Current Spawner: No

— Training Configuration —

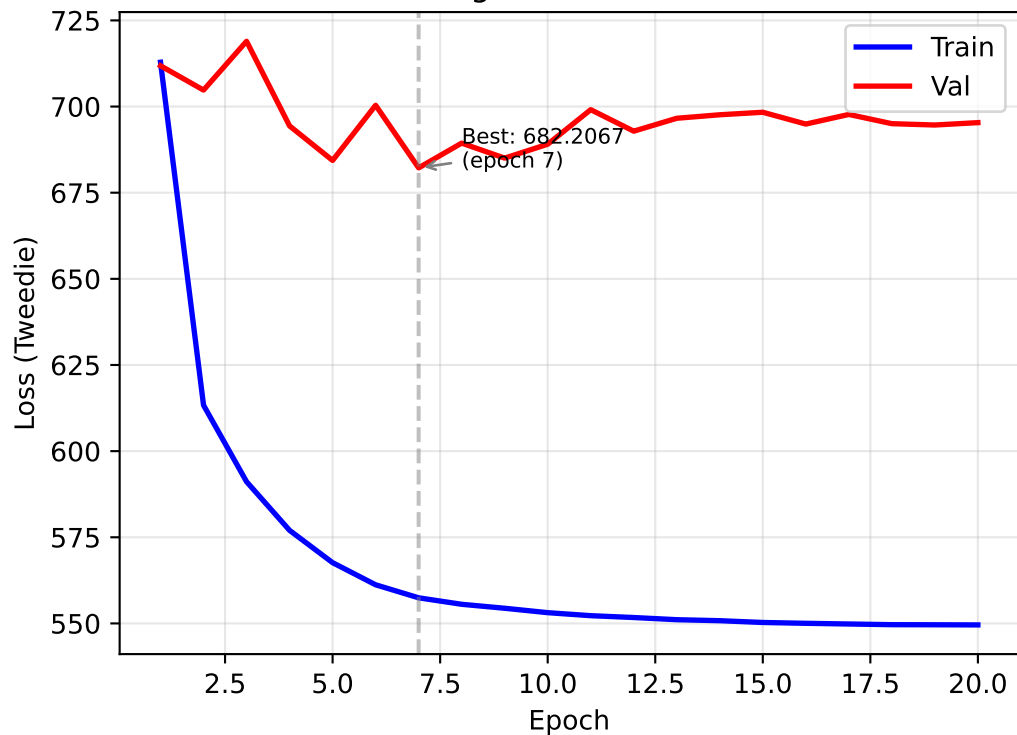
epochs_run: 20 best_val_epoch: 7
best_val_loss: 682.2067 final_train_loss: 549.5645
bias_correction (MSE): 1.0000
include_current_spawner: False

— Model Architecture —

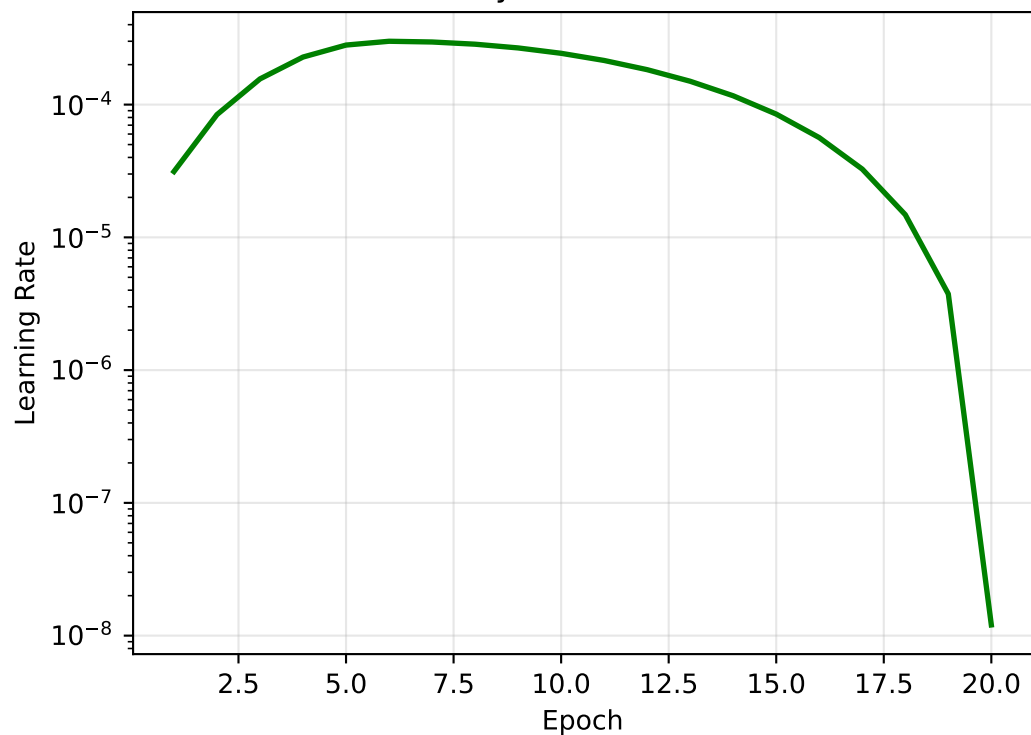
embed_dim: 128 num_heads: 4 num_layers: 3
d_ff: 512 dropout: 0.2 patch_size: 5
train/val/test years: 24/8/4

Training Curves: REAL | Lag 0 | Tweedie | NO Spawner(t)

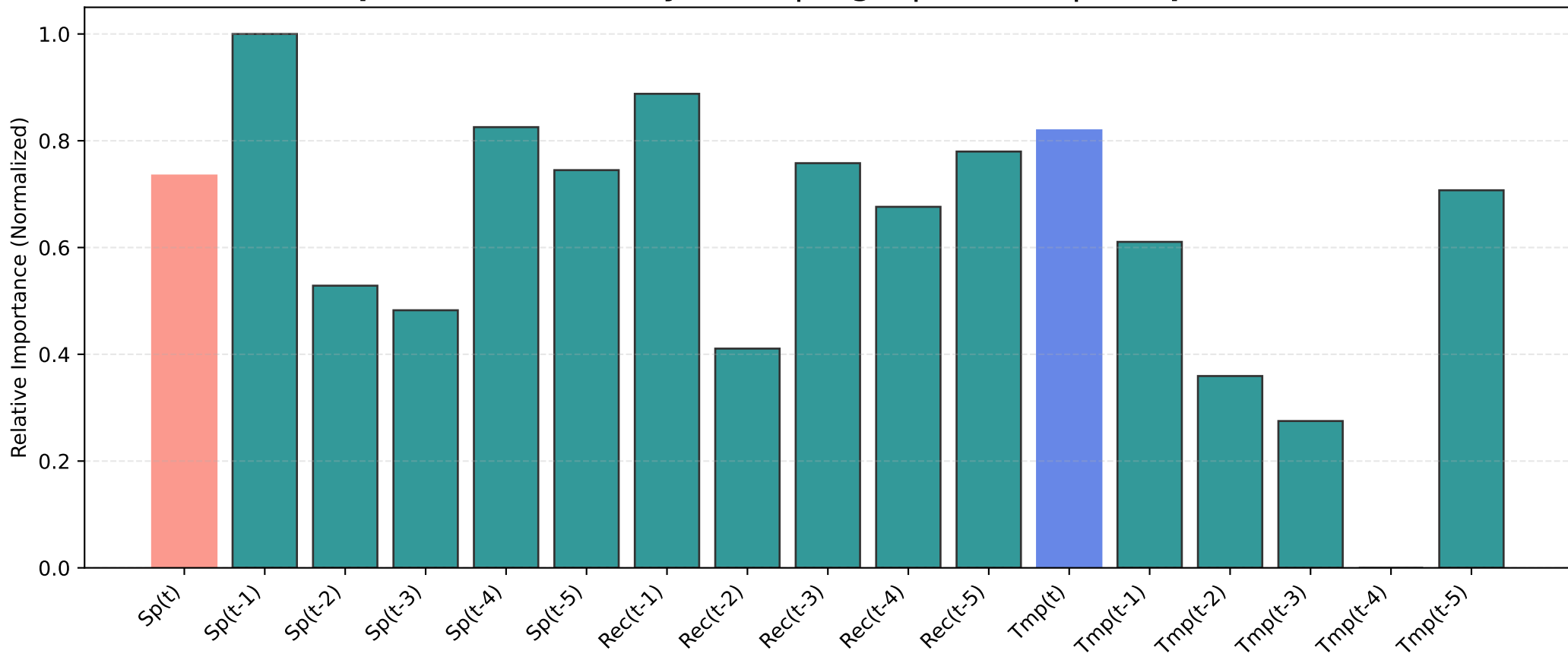
Training & Validation Loss



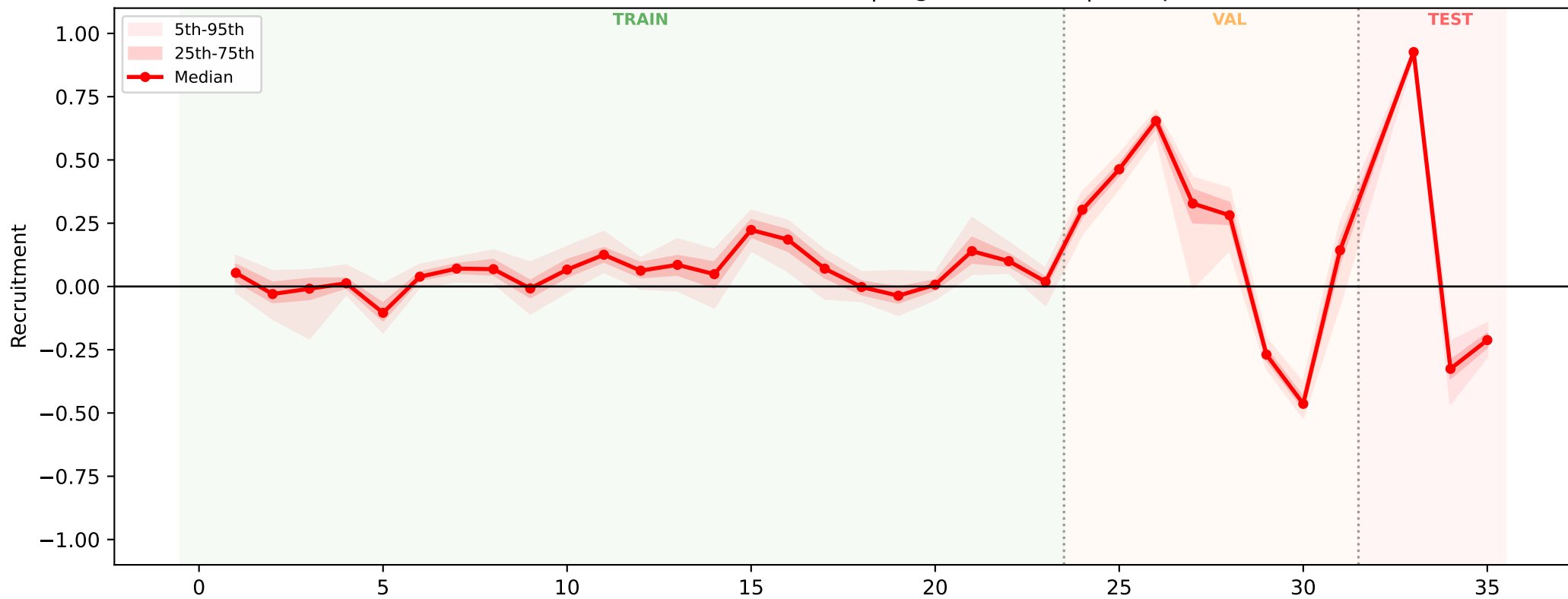
OneCycleLR Schedule



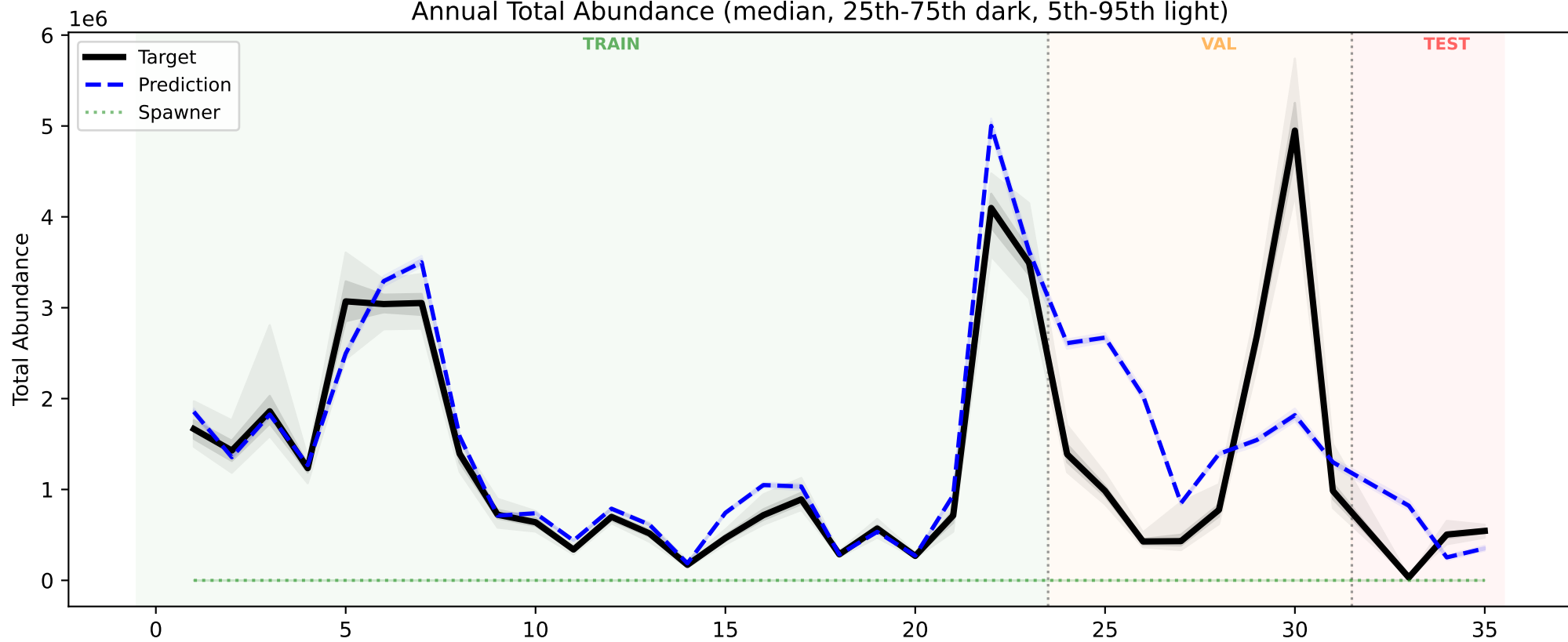
Input Channel Priority: REAL | Lag 0 | Tweedie | NO Spawner(t)



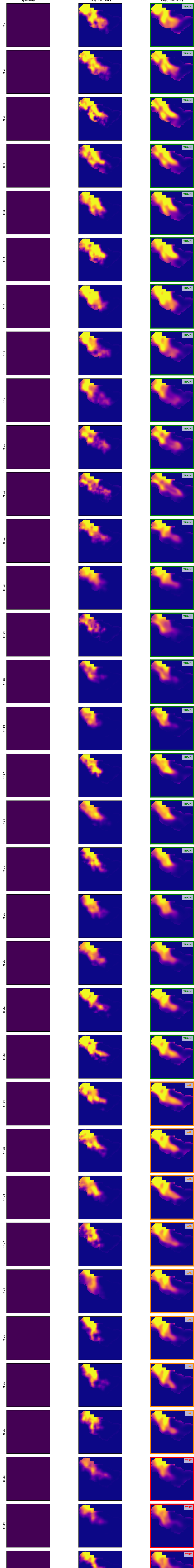
Annual Recruitment Deviations: REAL | Lag 0 (Tweedie) | NO Spawner(t)



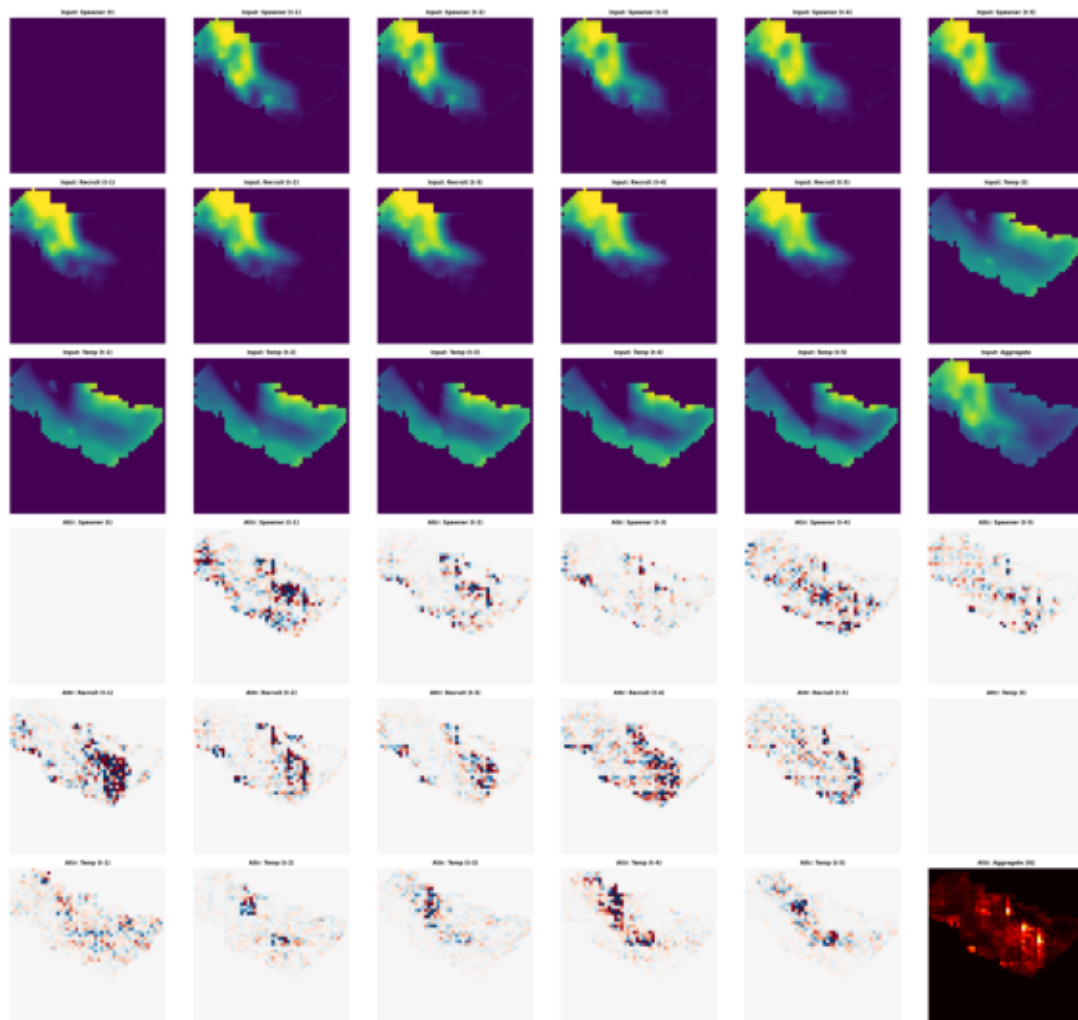
Annual Total Abundance (median, 25th-75th dark, 5th-95th light)



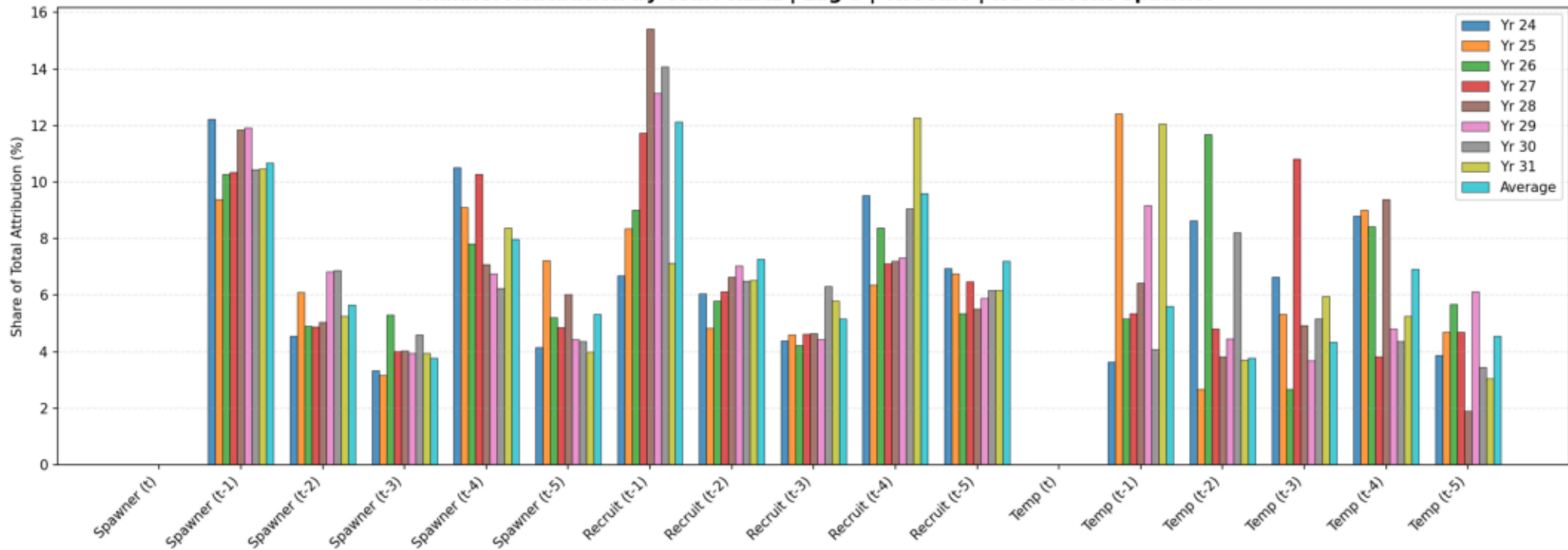
Spatial Trajectory: REAL | Lag 0 | Tweedie | NO Spawner(t)



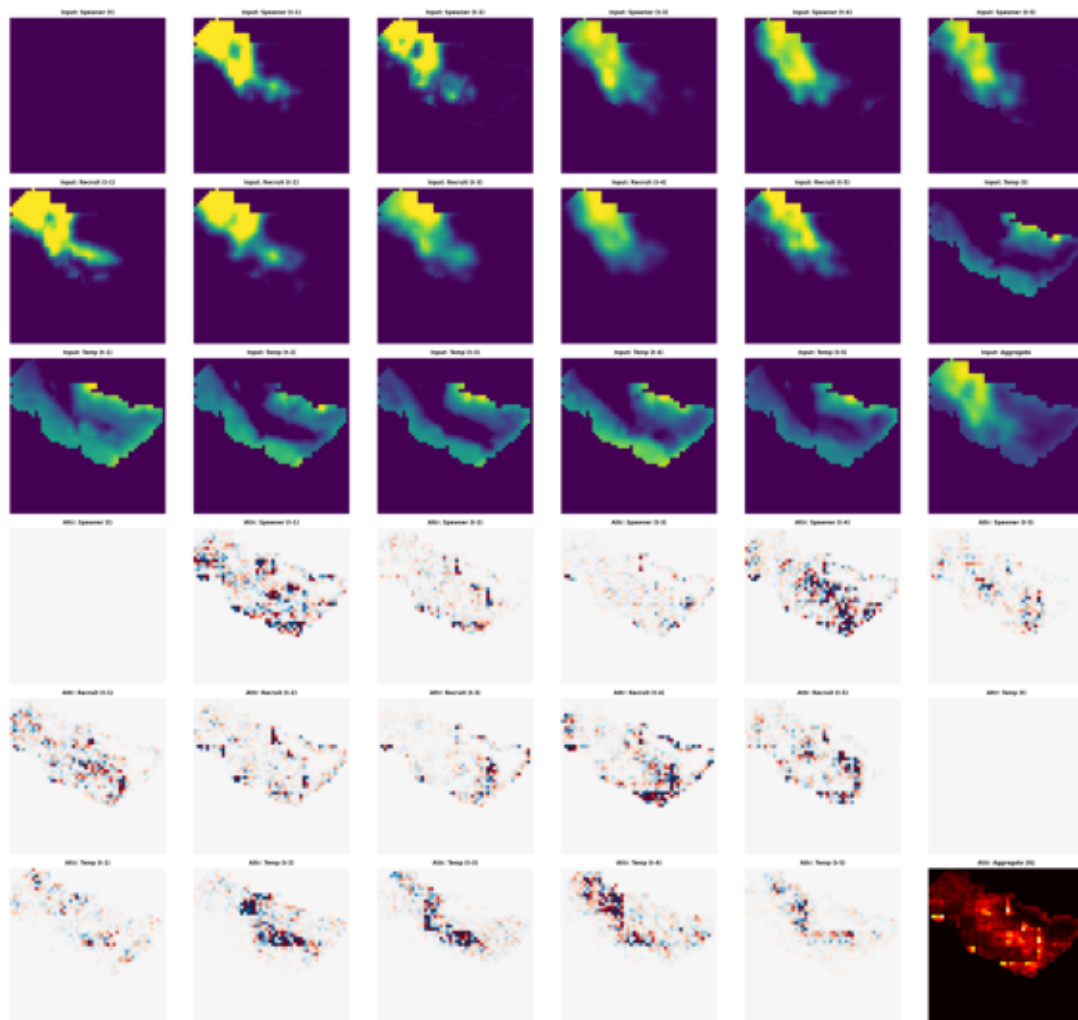
Year AVG -- REAL | Lag 0 | Tweedie | NO Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



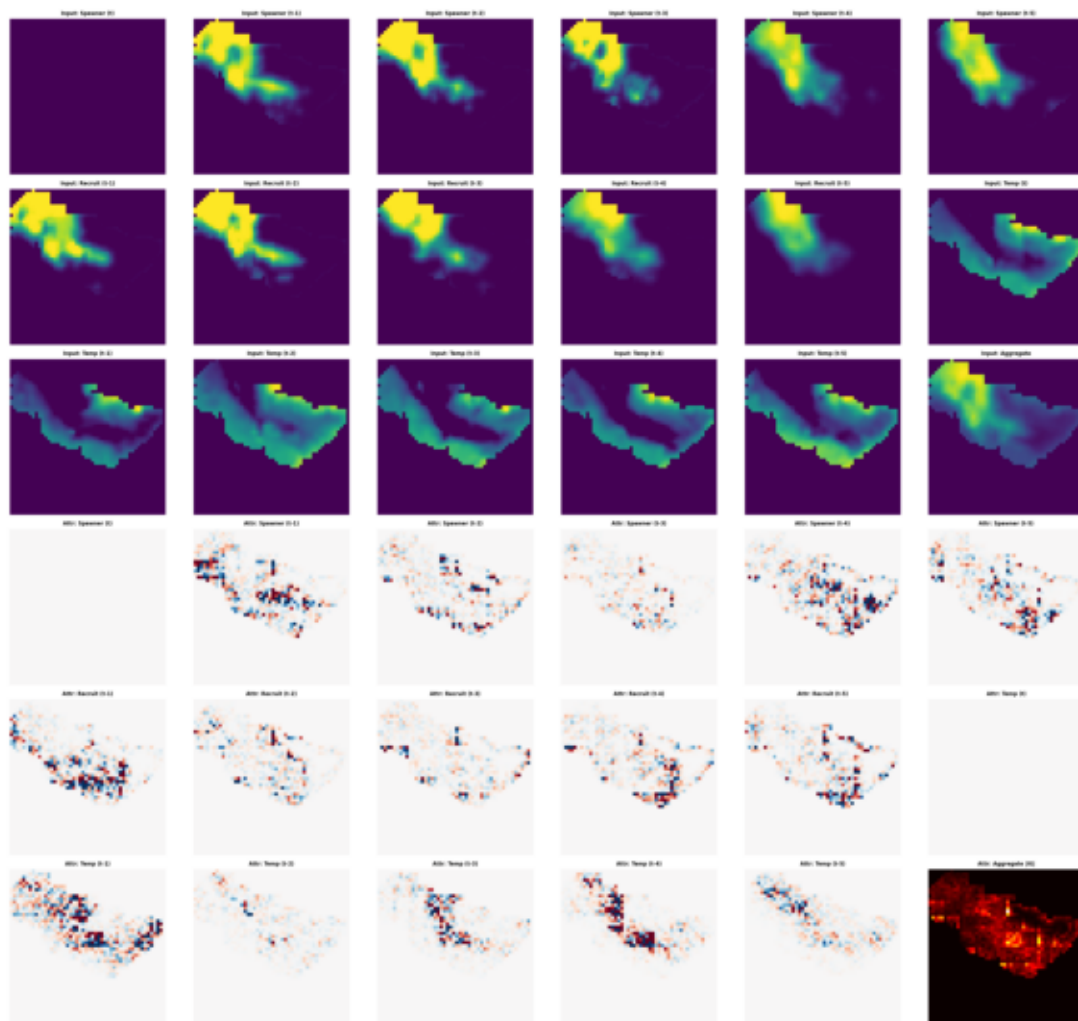
Channel Attribution by Year: REAL | Lag 0 | Tweedie | NO Current Spawner



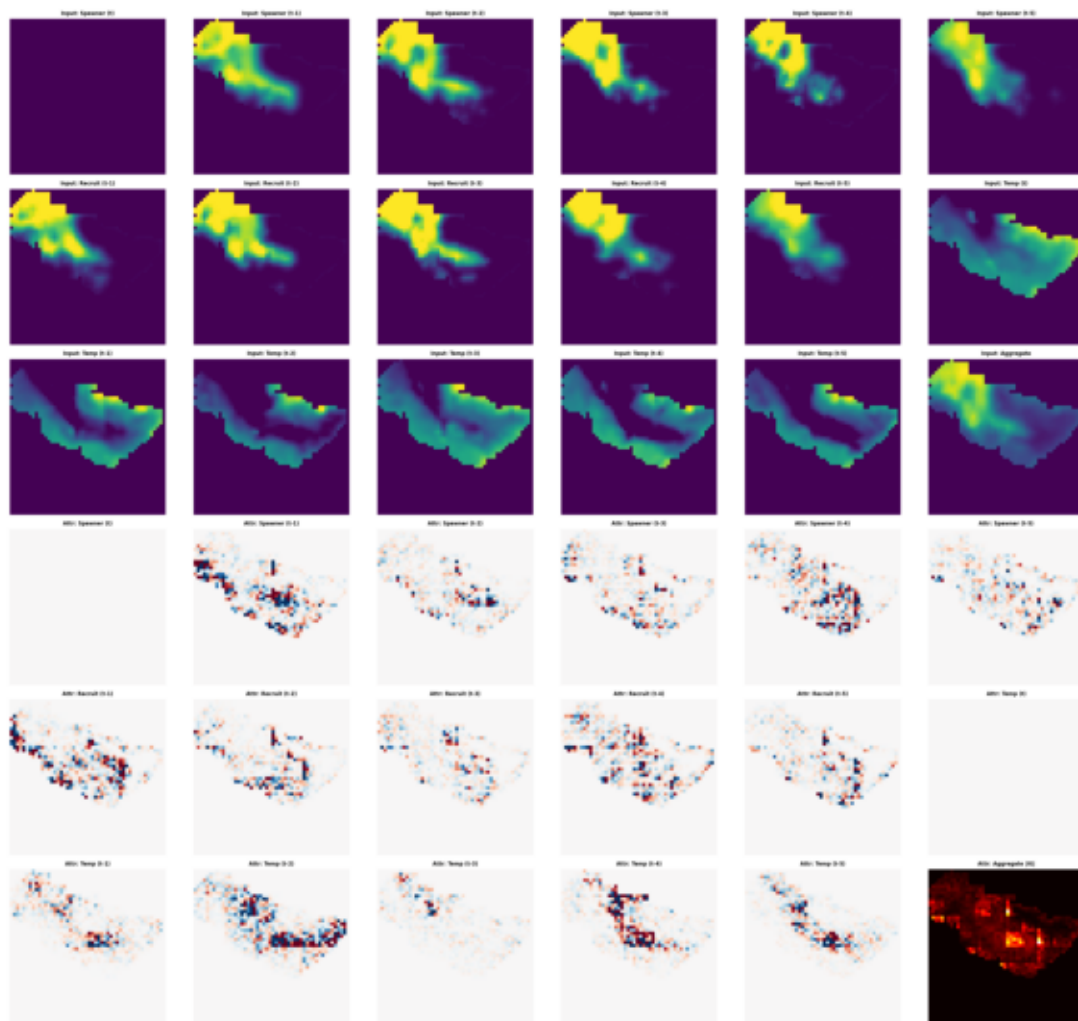
Year 24 -- REAL | Lag 0 | Tweedie | NO Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



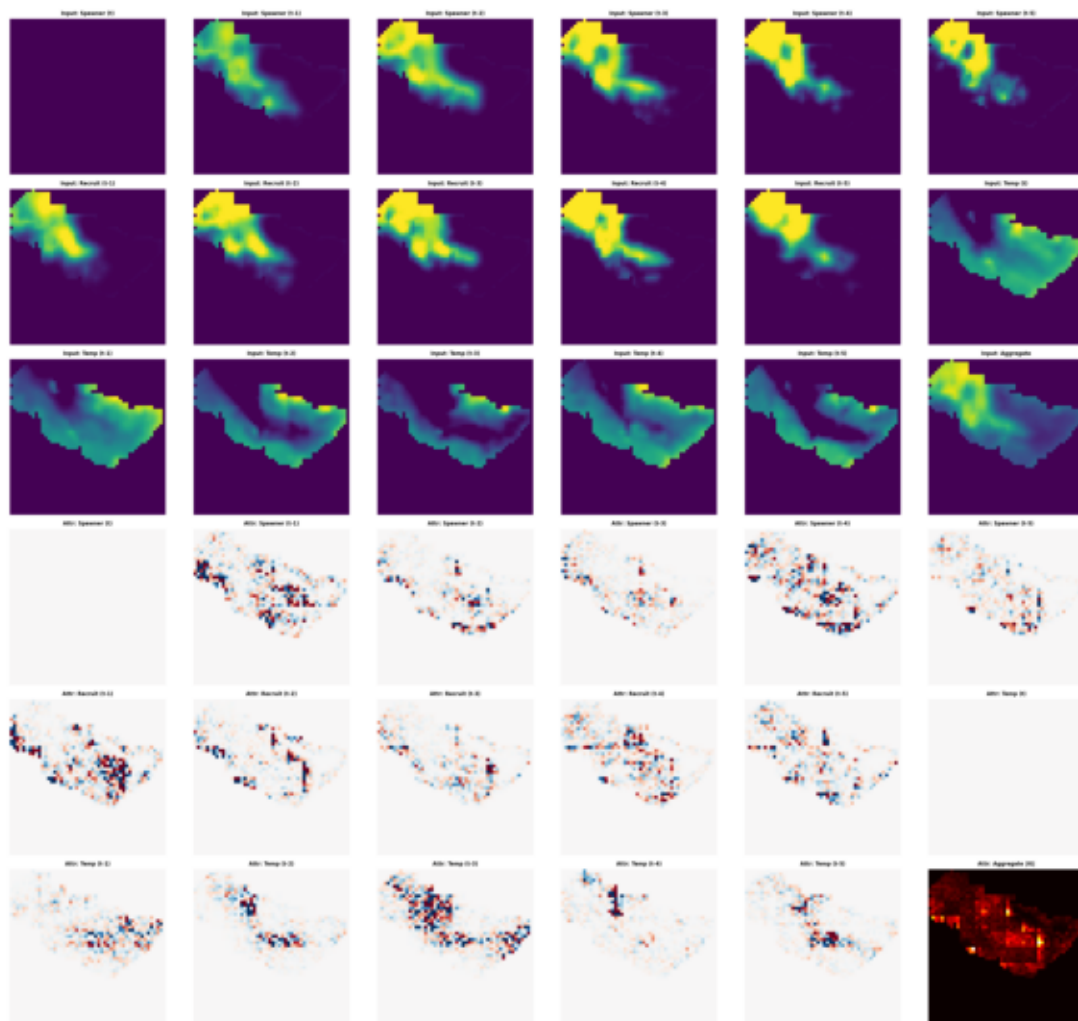
Year 25 -- REAL | Lag 0 | Tweedie | NO Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



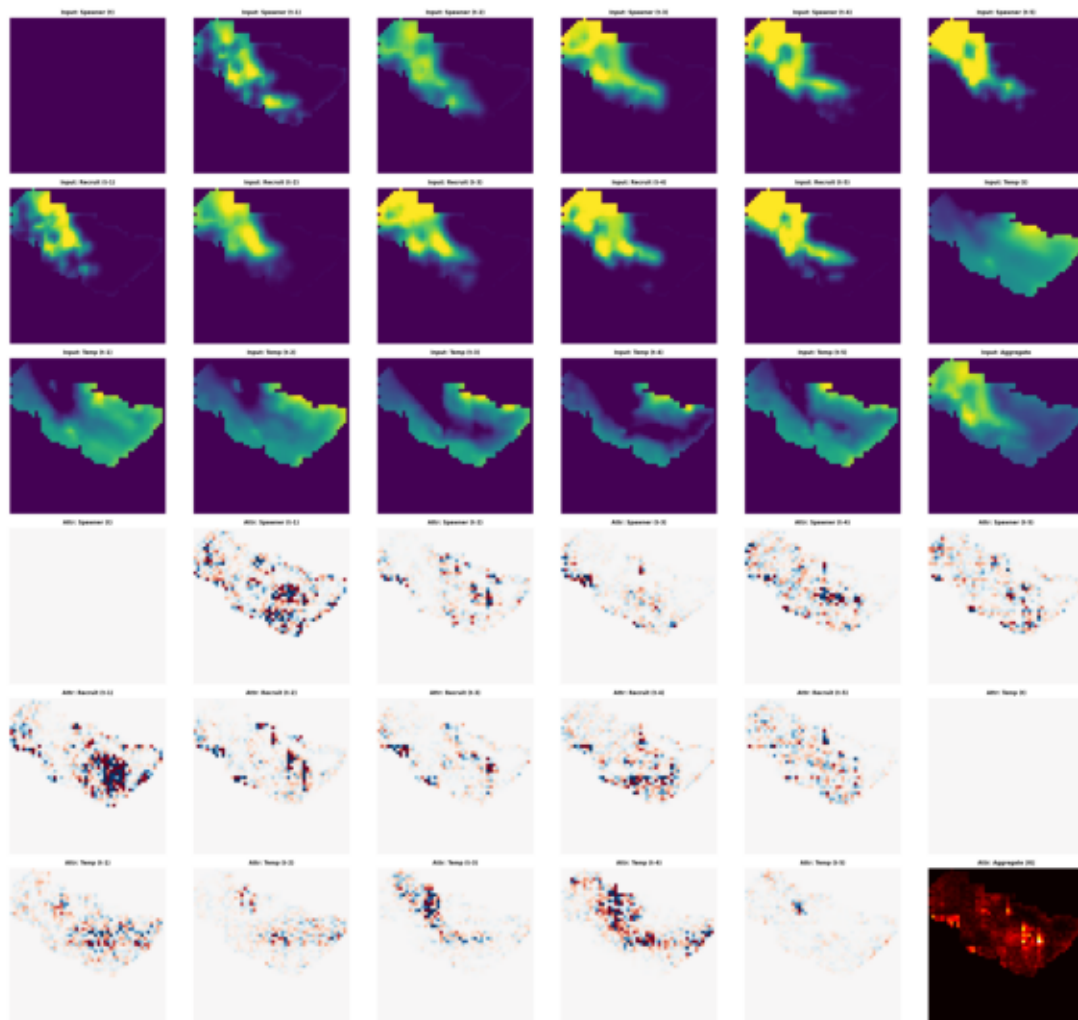
Year 26 — REAL | Lag 0 | Tweedie | NO Current Spawner
Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



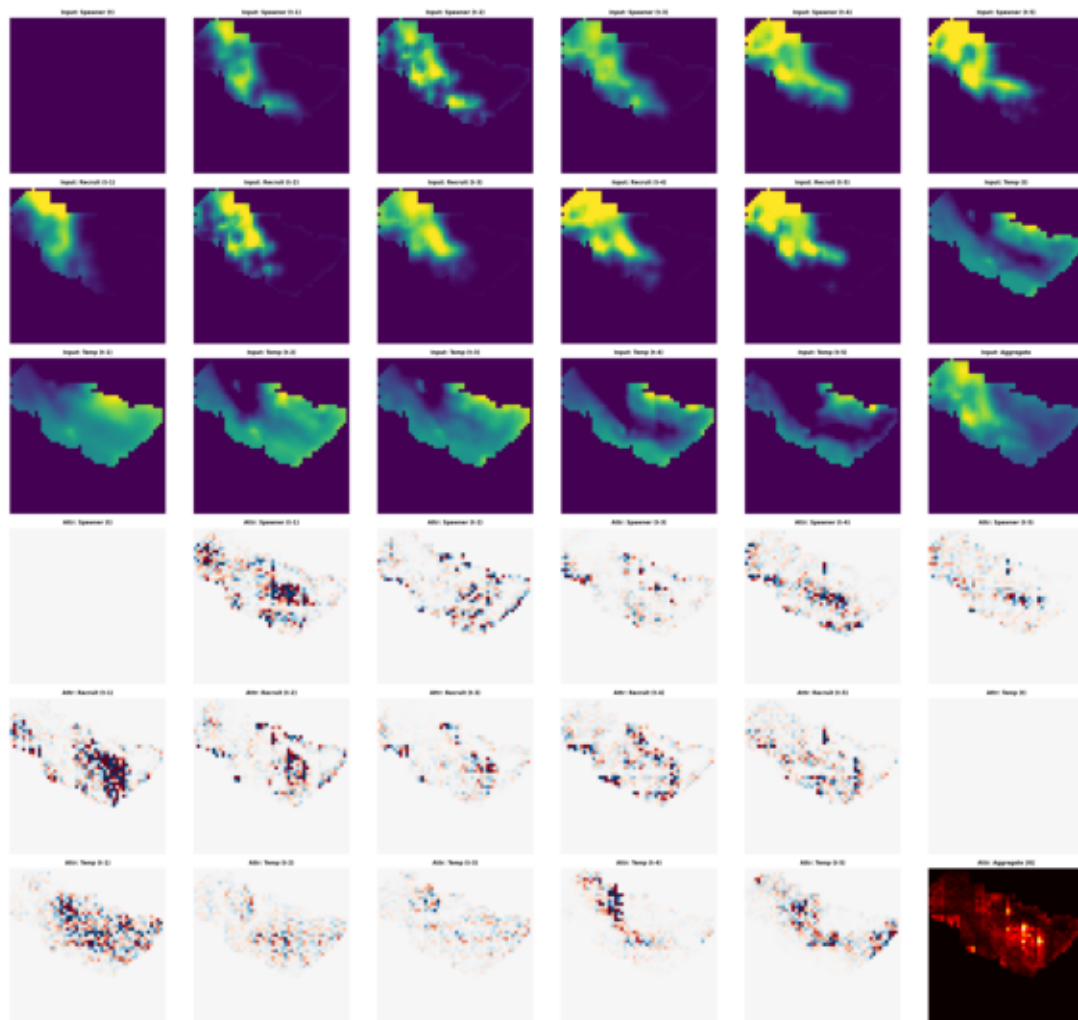
Year 27 -- REAL | Lag 0 | Tweedie | NO Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



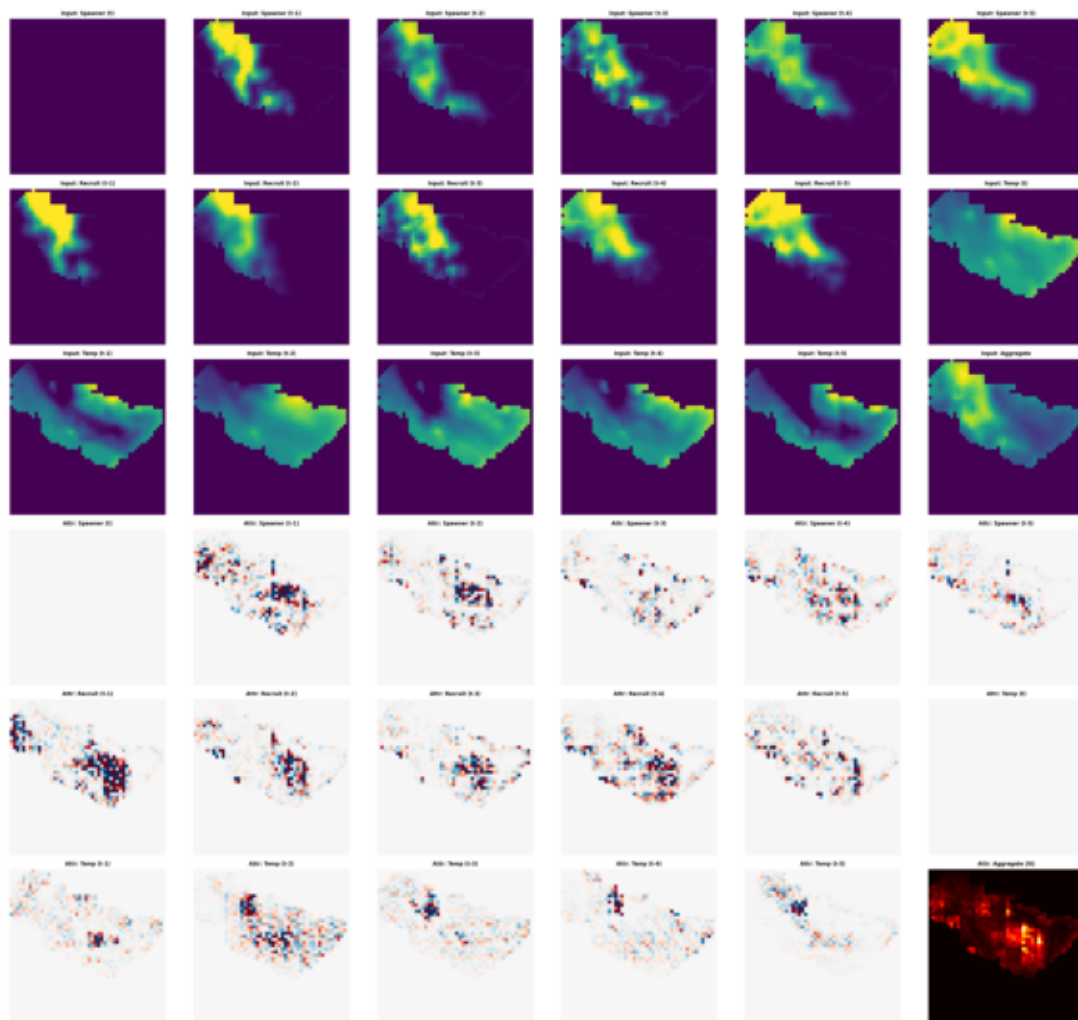
Year 26 -- REAL | Lag 0 | Tweedie | NO Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



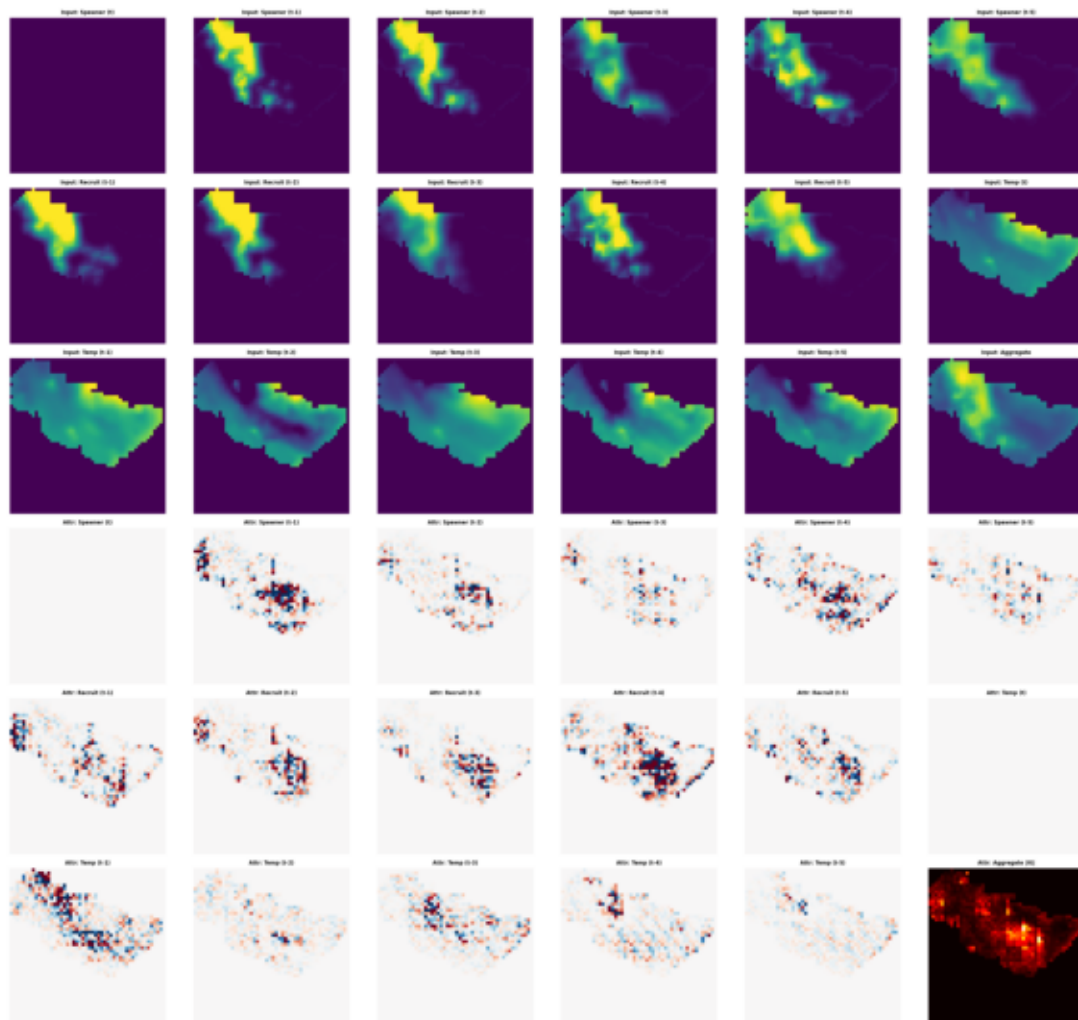
Year 29 -- REAL | Lag 0 | Tweedie | NO Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 30 -- REAL | Lag 0 | Tweedie | NO Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 31 -- REAL | Lag 0 | Tweedie | NO Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Summary Statistics: REAL | Lag 0 | Tweedie | NO Spawner(t)

Phase	Pred_Mean	Tgt_Mean	Pred_SD	Pred_P50	Pred_P95	Abundance_Capture	MAE	Spearman
TRAIN	1673.0	1542.8	6107.0	0.0	10054.8	1.084	489.5	0.939
VAL	2002.8	1801.5	5195.9	0.0	14159.4	1.112	2052.8	0.838
TEST	535.7	406.3	2191.0	0.0	3012.3	1.318	547.2	0.819

Detailed Metrics (R-style): REAL | Lag 0 | Tweedie | NO Spawner(t)

Phase	RMSE	RMSE_unbias	Bias	Bias_%	Spearman	Pearson	Pct_Error	Zero_F1	TopK_Acc	TopK_Jaccard	Quant_Err	Skill_MSE	Skill_MAE
TRAIN	1227.4	1204.7	130.2	3.1%	0.946	0.977	11.5%	0.944	0.915	0.846	0.36	0.911	0.829
VAL	6247.2	6072.2	201.3	5.5%	0.841	0.599	87.4%	0.882	0.609	0.454	1.01	-2.943	-0.037
TEST	2315.8	2263.4	129.4	4.3%	0.826	0.682	848.4%	0.917	0.720	0.582	0.76	-460.324	-4.964